



Day 6 Task - File Signatures, Deleted Files & Data Carving



Trios cyber • 9 Sept (Edited 11 Sept)

100/100 | Due 10 Sept, 11:00

Practical Assignment: File Signatures, Deleted Files & Data Carving

Dear Students,

For this practical assignment, you are required to examine file signatures and understand how deleted or residual data may be identified and recovered from digital storage.

The objective of this exercise is to understand how file type identification, deleted-file examination and data carving can assist in recovering potentially relevant digital evidence.

Part 1: File Signatures

Select at least 5 files from the forensic image created in Task 1.

For each file:

1. Identify the file extension.
2. Examine the file type/signature information available.
3. Compare the apparent file extension with the identified file type.
4. Record your observations.

Explain why a file extension alone should not always be treated as proof of the actual file type.

Part 2: Examine File Headers

Use an appropriate forensic or hex-viewing tool to examine the beginning of selected files.

Where possible, identify recognizable header/signature information.

Select at least 2 file types and document:

- File type
- File extension
- Signature/magic number
- Observed header information



Take screenshots of the relevant views.

Part 3: Deleted File Examination

Using a suitable test image containing deleted files:

1. Locate deleted-file entries, where available.
2. Examine the information associated with the deleted files.
3. Determine whether the underlying data appears to remain recoverable.
4. Document your observations.

Do not intentionally delete files from your Task 1 evidence.

Part 4: Data Carving

Using an approved forensic tool, perform a basic data-carving exercise.

Document:

- Source image
- File type/signature searched for
- Carving/recovery process
- Recovered files, where available
- Recovery location
- Any limitations or incomplete results

Take screenshots of the process and recovered results.

Part 5: Unallocated & Residual Data

Examine the available unallocated space, where applicable.

Explain:

- What unallocated space is
- Why deleted information may remain there
- How file carving can locate information without relying on normal directory structures
- Why recovery results must be examined carefully

Part 6: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. File-signature examination
4. Header/signature examples
5. Deleted-file examination
6. Data-carving procedure
7. Recovered results, where available
8. Unallocated-space observations
9. Screenshots



10. Conclusion

For every recovered item, clearly state whether it was directly observed/recovered or inferred from the available evidence.

Submission Requirements

Submit ONE file only:

- PDF (.pdf) OR
- Microsoft Word document (.docx)

Important Notes

Use only an authorized test image for deleted-file recovery and data-carving activities.

Do not intentionally delete or alter files in the forensic image created in Task 1.

Recovery results may contain incomplete, fragmented or unrelated data. Document what was actually observed.

Expected Learning Outcome

After completing this practical, you should be able to:

1. Explain file signatures and magic numbers.
2. Compare file extensions with actual file types.
3. Understand how deleted files may remain recoverable.
4. Explain the purpose of data carving.
5. Examine unallocated space for residual information.
6. Document recovered evidence and limitations.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

Please read the instructions carefully before beginning the assignment.

Your work

Marked

Day-6.pdf

PDF



+ Add or create

Resubmit

